

GenAI Hackathon

Intelligent Meeting Intelligence & Escalation Tracking System

Background

Modern organizations operate through a constant stream of meetings — spanning engineering, product, operations, support, HR, and leadership. Yet the valuable intelligence embedded within these meetings rarely survives in an actionable form.

Critical information — escalations, blockers, project risks, ownership transfers, action items, deadlines, and key decisions — is routinely buried inside lengthy transcripts, loosely formatted summaries, or informal chat threads. By the time these insights are needed, they are difficult to find, impossible to query, and entirely disconnected from related context.

The consequences are significant:

- Escalations go untracked until they become crises
- Project risks accumulate invisibly across teams
- Accountability is lost when ownership is undocumented
- Historical context is inaccessible when decisions need revisiting
- Leadership lacks real-time visibility into organizational health

Generative AI now makes it possible to change this — by transforming unstructured meeting content into structured, searchable, and interconnected organizational knowledge in real time.

Problem Statement

Design and build an AI-powered Meeting Intelligence System that ingests raw meeting content — summaries, transcripts, or discussions — and automatically extracts, structures, and interconnects the organizational intelligence hidden within. where the endpoint will be created to provide the example meeting conversation as an input

The system must intelligently surface:

- Projects & initiatives under discussion
- Action items and task ownership
- Escalations, blockers, and unresolved issues
- Risks, dependencies, and cross-team impacts
- Stakeholders, owners, and accountable parties
- Timelines, deadlines, and commitments
- Priority levels and urgency signals

- Key decisions and their rationale
- Follow-up actions and next steps

Once extracted, this intelligence must be:

- Stored in a structured, queryable format
- Contextually linked — connecting people, tasks, risks, and projects
- Accessible through natural language queries or API interfaces
- Presented in a way that drives real organizational decisions

Objective

Convert unstructured meeting discussions into actionable business intelligence using Generative AI — enabling teams to query their own organizational history, track accountability, and proactively manage risks without manual effort.

Illustrative Scenario

Input — Raw Meeting Summary:

“The payment integration is delayed because the Vendor API is unstable. Rahul will coordinate with the backend team before Friday. If this issue continues, it may impact the Phase-2 release. Priya escalated the concern to leadership.”

Expected AI Output:

- **Project:** Payment Integration
- **Blocker:** Vendor API instability
- **Owner:** Rahul
- **Deadline:** Friday
- **Risk:** Delay in Phase-2 release
- **Escalation Raised By:** Priya
- **Teams Involved:** Backend Team
- **Priority:** High

Core Capabilities to Build

1. Meeting Input Processing

The system should accept meeting content in multiple forms:

- Structured or free-form meeting summaries
- Raw text transcripts

- Uploaded document files

2. Intelligent Information Extraction

Using AI and NLP, the system should reliably identify and extract:

- Tasks and action items
- Escalations and unresolved blockers
- Risks and inter-team dependencies
- Key decisions and open questions
- Deadlines and time-bound commitments
- Ownership and stakeholder mapping

3. Structured Knowledge Storage

Extracted information must be stored in a structured, queryable format. Participants may use relational databases (SQL), document stores (NoSQL), or graph databases such as Neo4j to model relationships like:

- Employee → Assigned_To → Task
- Project → Has_Blocker → Issue
- Escalation → Raised_By → Person

4. Conversational Querying

Users should be able to ask natural language questions and receive precise, context-aware answers. Examples:

- *“What are the current unresolved escalations?”*
- *“Which projects are at risk this week?”*
- *“Show all pending tasks assigned to Rahul.”*
- *“Which meetings discussed Vendor API issues?”*
- *“List all high-priority blockers across teams.”*

5. Organizational Insight Generation

Beyond individual queries, the platform should surface meaningful patterns across meetings:

- Escalation trends over time
- Project health and risk status
- Pending action items and accountability gaps
- Team workload distribution
- Dependency and risk mapping

Bonus Opportunities

Teams that complete the core system may explore advanced capabilities to differentiate their solution:

- Multi-agent orchestration for parallel extraction workflows
- Semantic search using vector embeddings (ChromaDB, FAISS)
- Risk severity scoring and prioritization engine
- Duplicate escalation detection across meetings
- Sentiment analysis on meeting tone and urgency
- Auto-generated follow-up emails or meeting action reports
- Slack or Microsoft Teams notification integration
- Real-time dashboards for leadership visibility
- Voice-based meeting ingestion via speech-to-text

Expected Outcome

The final solution should demonstrate how Generative AI can transform raw, unstructured meeting discussions into a structured, searchable, and interconnected organizational intelligence system — one that actively tracks tasks, escalations, risks, ownership, and operational health across an entire organization.